Grade-Skewed Domain Adaptation via Asymmetric Bi-Classifier Discrepancy Minimization for Diabetic Retinopathy Grading

Yuan Ma, Yang Gu, *Member, IEEE*, Shuai Guo, Xin Qin, Shijie Wen, Nianfeng Shi, Weiwei Dai, and Yiqiang Chen, *Senior Member, IEEE*

***Abstract*— Diabetic retinopathy (DR) is a leading cause of preventable low vision worldwide. Deep learning has exhibited promising performance in the grading of DR. Certain deep learning strategies have facilitated convenient regular eye check-ups, which are crucial for managing DR and preventing severe visual impairment. However, the generalization performance on cross-center, cross-vendor, and cross-user test datasets is compromised due to domain shift. Furthermore, the presence of small lesions and the imbalanced grade distribution, resulting from the characteristics of DR grading (e.g., the progressive nature of DR disease and the design of grading standards), complicates image-level domain adaptation for DR grading. The general predictions of the models trained on grade-skewed source domains will be significantly biased toward the majority grades, which further increases the adaptation difficulty. We formulate this problem as a grade-skewed domain adaptation challenge. Under the grade-skewed domain adaptation problem, we propose a novel method for image-level supervised DR grading via Asymmetric Bi-Classifier Discrepancy Minimization (ABiD). First, we propose optimizing the feature extractor by minimizing the discrepancy between the predictions of the asymmetric bi-classifier based on two classification criteria to encourage the exploration of crucial features in adjacent grades and stretch the distribution of adjacent grades in the latent space. Moreover, the classifier difference is maximized by using the forward and inverse distribution compensation mechanism to locate easily confused instances, which avoids pseudo-label bias on the target domain. The experimental results on two public DR datasets and one private DR dataset demonstrate that our method outperforms state-of-the-art methods significantly.**

***Index Terms*— Asymmetric bi-classifier discrepancy, diabetic retinopathy grading, grade-skewed domain adaptation, adversarial learning, transfer learning.**

I. Introduction

DIABETIC Retinopathy (DR) is a common complication of diabetes and a leading cause of preventable low vision worldwide [1]. According to the IDF Diabetes Atlas [2], the global prevalence of diabetes among adults exceeded 10% in 2021. The more serious problem is that according to the forecast, the diabetic population will continue to experience substantial growth in the future. Accurately grading the severity of DR is crucial for effective management and treatment of the condition, as 90% of severe vision loss can be prevented through routine ophthalmic examinations coupled with effective treatment [3]. Over the past few years, deep learning-based methods [4], [5] have shown promising results in automating DR grading, providing a reliable and efficient alternative to traditional grading systems. And among these methods, the pixel-level supervision method based on lesion identification stands out for its ability to achieve remarkably high accuracy.

Unfortunately, the pixel-level annotation of lesions on retinal images is costly [6], making image-level supervised grading crucial for assisting clinical diagnosis and treatment planning in DR grading. Existing image-level supervised methods conduct experiments on publicly available datasets that adhere to the assumption of independent and identically distributed (i.i.d.), where the datasets are randomly partitioned into training and test sets. However, cross-domain grading in practical applications has rarely been explored.

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Yuan Ma, Yang Gu, Shuai Guo, Xin Qin, Shijie Wen, and Yiqiang Chen are with the Institute of Computing Technology and Beijing Key Laboratory of Mobile Computing and Pervasive Device, Institute of Computing Technology, Chinese Academy of Sciences, Beijing 100045, China, and also with the School of Computer Science and Technology, University of Chinese Academy of Sciences, Beijing 101408, China (e-mail: mayuan20z@ict.ac.cn; guyang@ict.ac.cn; guoshuai20g@ict.ac.cn; qinxin@ict.ac.cn; wenshijie20s@ict.ac.cn; yqchen@ict.ac.cn).

Nianfeng Shi is with Luoyang Institute of Science and Technology, Luoyang, Henan 471023, China (e-mail: shinf@lit.edu.cn).

Weiwei Dai is with Changsha Aier Eye Hospital, Changsha, Hunan 410015, China (e-mail: daiweiwei@aierchina.com).

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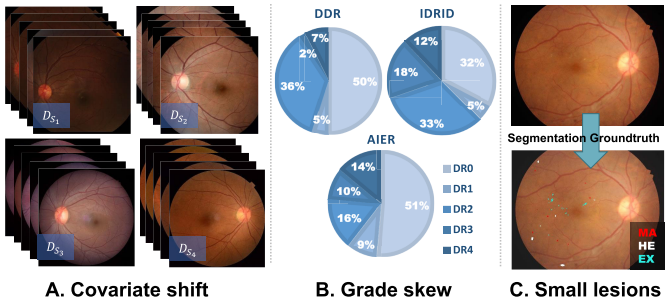


Fig. 1. Cross-center diabetic retinopathy grading is a typical grade-skewed domain adaptation (GSDA) problem. We highlight three main challenges in GSDA: covariate shift (in the four domains of the Aier dataset), grade skew, and small inter-grade differences (small lesions).

In light of the existing circumstances, the image-level supervised DR grading adaptation task is still challenging [7]. As illustrated in Fig. 1, this difficulty can be attributed to three principal causes:

Firstly, during the practical application of the trained model for DR grading, the test data is often significant disparity from the training data in the equipment, scenario and region. As illustrated in Fig. 1(A), the fundus images obtained in the four real hospitals are visually different. It may degrade the classification performance due to the shift in domains. We attribute the disparity between cross-center, cross-vendor and cross-user domain to covariate shift $P_S(\mathbf{x}) \neq P_T(\mathbf{x})$.

Secondly, the DR grades are not evenly distributed. As shown in Fig. 1(B), the highly skewed grade distribution exists with both the publicly available datasets and the actual datasets available in hospitals. The imbalanced data distribution makes the classifier biased towards these majority grades naturally. The label distribution of DR grades $P(y)$ is heavily skewed. Moreover, the different label distributions between source domain and target domain are defined as label shift, that is $P_S(y) \neq P_T(y)$, $P_S(\mathbf{x}|y) \neq P_T(\mathbf{x}|y)$.

Thirdly, as demonstrated in Fig. 1(C), the lesion features difference between the adjacent grades was not significant, which further increases the possibility of misclassification and make it easier for minority grades to be divided into the neighboring majority grades. For example, according to the International Clinical Diabetic Retinopathy Scale [3], the clinical sign of mild DR (DR1) is microaneurysms, which typically appear as subtle color variations of a few pixels on fundus images. At the same time, the proportion of DR1 within the population is extremely low. As it turns out, the instances of DR1 are often misclassified as no apparent retinopathy (DR0) or moderate nonproliferative diabetic retinopathy (DR2).

By formulating the problem with the above three characteristics as ***Grade-Skewed Domain Adaptation*** (GSDA), we aim to address this practical challenge of both covariate and label shift between source and target domains with small inter-grade feature differences but large inter-grade quantitative differences. This is a real-world problem that has not been fully explored in the literature, but is very common in applications such as medical grading.

To alleviate the grade-skewed domain adaptation problems, we propose a novel method called Asymmetric Bi-Classifier Discrepancy Minimization (ABiD), which introduces a new framework and an innovative adversarial training approach.

For model architecture, we employ a shared feature extractor whose outputs are fed into an asymmetric bi-classifier system based on Bayes-optimal balanced risk minimization and empirical risk minimization strategies. This design ensures that the model can effectively identify the confusing points near minority grades. Building on this, we train the feature extractor and classifiers in an adversarial manner, allowing the model to learn essential representations that capture the intrinsic characteristics of the data. Specifically, we introduce a forward and inverse compensation mechanism to amplify the differences between the classifiers: forward distribution compensation for the general classifier and inverse compensation for the Bayes-optimal classifier, ensuring that the process is unaffected by label distribution. This mechanism allows our method to identify confusion points caused by grade imbalance, small lesions, and covariate shift, whereas traditional DA methods typically detect the points caused only by covariate shift. After thoroughly identifying these confusion points, we train the feature extractor to learn essential distinguishing features that do not produce discrepancies between the asymmetric classifiers. Unlike methods that adjust classifiers to shift predictions toward minority classes, our method focuses on finding essential features to enhance model performance.

Our main contributions can be summarized as follows:

- We define a new and applicable challenge of Grade-skewed Domain Adaptation (GSDA). To the best of our knowledge, this is the first work for GSDA to tackle covariate shift and label shift simultaneously in the grade-skewed datasets.
- We propose a novel framework for grade-skewed domain adaptation, which includes the asymmetric bi-classifier based on different classification strategies respectively.
- We propose a new adversarial training approach that utilizes two forward and inverse compensate for enhancing the disparity between bi-classifier near the decision boundaries, and then forces the feature extractor to generate features that minimize the disparity.
- The experiments on one toy dataset (inter-twining moons), two public datasets (*IDRiD, DDR*) and one private dataset (*Aier*) with 21 cross-domain tasks demonstrate the effectiveness of the proposed method.

II. Related Works

A. Long-Tail Distribution and Grade-Skewed Distribution

Both the long-tail (LT) distribution [8], [9] and the grade-skewed distribution belong to the imbalanced distribution, and the class re-balancing [10], [11] methods can improve the LT and GSDA problem. On the other hand, we present several clarifying observations on grade-skewed distribution, distinguishing it from the long-tail distribution: The long-tail distribution arises due to the existence of minority classes in datasets collected naturally, whereas the grade-skewed distribution can occur even with a few of grades. The variation in the number of instances among grades may be attributed to

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factors such as the grading criteria design, disease severity, patient populations, or the average duration of illness grade. Consequently, in this context, the target domain generally encounters a skewed distribution similar to that of the source domain. Furthermore, given the progressive nature of disease staging, there are likely to be only subtle differences in lesion characteristics between adjacent grades, making them more challenging to distinguish.

B. Unsupervised Domain Adaptation

Unsupervised domain adaptation (UDA) has emerged as a promising way to transfer knowledge from a labeled source domain to an unlabeled target domain.

1) Domain Adversarial Learning: Adversarial learning-based DA methods aims to address this issue by introducing an adversarial training process that aligns the distributions of the source and target domains. The pioneering attempt is DANN [12], which achieved distribution alignment by adding an adversarial loss so that the domain classifier cannot distinguish the features from the source or target domain.

Dual-classifier adversarial methods [13], [14] maximize classifier discrepancies based on decision boundaries trained by the model. However, This approach often leads to biases when there are significant differences in grade frequencies. Both decision boundaries tend to align with majority grades, neglecting minority grade confusion points. This issue disrupts the feature extractor during adversarial training, especially when inter-grade differences are minimal, leading to error accumulation. Improvements like BCDM [15], which uses CDD instead of the $L1$ distance in MCD, increase distinguishability but worsen minority class discrimination in imbalanced grade scenarios. Our method addresses these issues by leveraging discrepancies between predictions based on different criteria to accurately locate confusion points near minority grades. Forward and backward distribution compensation maximizes classifier differences, unaffected by label distribution biases. This new method allows us to identify confusion points from grade imbalance, small lesions, and covariate shifts.

2) Joint Distribution Adaptation: Closely related to GSDA, recent work [16] raises the issue of class-imbalanced domain adaptation (CDA) and indicating that data shift and label shift should be handled simultaneously for CDA problem, which is consistent with the goal of earlier work on joint distribution adaptation. The line of joint distribution adaptation [17], [18] in DA literature consider both conditional distribution shift and label distribution shift. By theoretically analyzing the lower bound on the joint risk, Zhao et al. [19] pointed out that when the label distributions of the source and target domains are significantly different, forcing the learning of invariant representations will degrade the performance of targets.

Joint Adaptation Network (JAN) [20] considered joint distribution adaptation for the first time in deep DA. Several works have explored methods such as leveraging committee consistency [21], using interpolated adversarial samples, and employing centroid alignment strategies [22]. Moreover, the latest works [23], [24], [25] introduce innovative strategies, such as adversarial training on penultimate activations and iterative adaptation paradigms. However, they may still encounter issues like early training errors or suboptimal label accuracy.

3) Applications in Medical Images: Medical imaging tasks often face significant variability due to differences in equipment, environment, and demographics, driving the development of domain adaptation methods. CDACM [26] used conditional adversarial learning to reduce domain gaps in pneumonia diagnosis. Pro-MDD [27] employed prototype learning to constrain feature outliers and align source and target domains. However, these approaches mainly address domain shifts without targeting class imbalance. Meanwhile, methods like CANet [6], BiRA-Net [28] and CABNet [7] have been proposed for DR grading to tackle imbalanced data distributions, but they struggle under broader domain shift scenarios [1]. Our method advances these DA methods by tackling GSDA, addressing the complexities of covariate and label shifts within grade-skewed data, which will be defined in Section III.

III. Methodology

A. Problem Definition and Notations

According to the International Clinical Diabetic Retinopathy Disease Severity Scale [3], the DR is generally divided into 5 grades (see Fig. 5 (bottom right) for details). We use DR0-DR4 to denote the grade of no apparent retinopathy (DR0), mild nonproliferative diabetic retinopathy (DR1), moderate nonproliferative diabetic retinopathy (DR2), severe nonproliferative diabetic retinopathy (DR3) and proliferative diabetic retinopathy (DR4), respectively. The GSDA problem in DR grading can be framed as a grading task for target domain with the same label set with sources $\mathcal{Y} = \{y_i\}_{i=0}^{4}$. The labeled source domain is denoted as $\mathcal{D}_S = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N_S}$ and unlabeled target domain is denoted as $\mathcal{D}_T = \{\mathbf{x}_i\}_{i=1}^{N_T}$, where $\mathbf{x}_i \in \mathcal{X}$ stands for input, $y_i \in \mathcal{Y}$ stands for the DR-i grade.

We introduce two asymmetric classifiers C_{ERM}, C_{BO} based on different classification criteria: C_{ERM} is based on the empirical risk minimization, while C_{BO} is based on the Bayes-optimal balanced risk minimization. The extracted features $\mathbf{z} \in \mathcal{Z}$, which serve as input for the classifier, are generated by the feature extraction function F. We define $\circ$ to be the function composition operator. Accordingly, the hypothesis family on the source and target domains can be defined as $\mathcal{H} := \{h(\cdot) = C \circ F(\cdot) : \mathcal{X} \xrightarrow{F} \mathcal{Z} \xrightarrow{C_{\text{ERM/BO}}} \mathcal{Y}_{\text{out}} \xrightarrow{softmax} \mathcal{Y}_{\triangle}\}$, where $\mathcal{Y}_{\triangle} := \{\boldsymbol{v} \in \Delta^4 : \|\boldsymbol{v}\|_1 = 1, v_i \geq 0 \text{ for } i = 0, 1, \ldots, 4\}$ is the softmax of classifier output $\mathcal{Y}_{\text{out}}$ and represents a 4-simplex. And we define the known term $\boldsymbol{\pi}_s$ as the grade priors of the source domain.

A replaceable augmentation module is denoted as $\mathcal{M}(\cdot)$, which applies transformations to an input $\mathbf{x}$, resulting in an augmented version $\widetilde{\mathbf{x}} = \mathcal{M}(\mathbf{x})$.

Comparison of CDA and GSDA: Compared to CDA, GSDA in the presence of data shift and label shift, adjacent grades with similar features are highly likely to have huge quantitative differences, which makes the GSDA problem more challenging. To formalize this difference, we introduce two key definitions:

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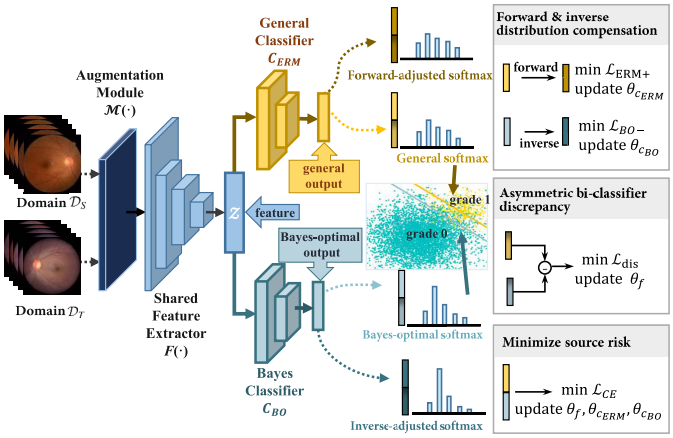


Fig. 2. Illustration of the Asymmetric Bi-Classifier framework for GSDA. The framework includes a shared feature extractor (blue), a general classifier (yellow), and a Bayes-optimal classifier (green).

Definition 1 (Label Distribution Discrepancy $\mathcal{C}$): Let Label distribution discrepancy matrix $\mathcal{C} \in \mathbb{R}^{V\times V}$. Then, label distribution discrepancy between two i^{th} grade and j^{th} grade is defined as:

$$\mathcal{C}(i, j) = |P(y = i) - P(y = j)| \tag{1}$$

Definition 2: Denoting the label space of the CDA problem as $\mathcal{Y}_{\text{CDA}} = \{y_i^{\text{C}}\}_{i=0}^{V}$ and that of the GSDA problem as $\mathcal{Y}_{\text{GSDA}} = \{y_i^{\text{G}}\}_{i=0}^{V}$, for all $\forall(\mathbf{x}_i^{\text{C}}, y_i^{\text{C}}), (\mathbf{x}_j^{\text{C}}, y_j^{\text{C}})$ and $\forall(\mathbf{x}_i^{\text{G}}, y_i^{\text{G}}), (\mathbf{x}_j^{\text{G}}, y_j^{\text{G}})$, $0 \le y_i < y_j < V$, we have

$$\min \frac{dist(F(\mathbf{x}_i^{\text{G}}), F(\mathbf{x}_j^{\text{G}}))}{\mathcal{C}(y_i^{\text{G}}, y_j^{\text{G}})} \ll \min \frac{dist(F(\mathbf{x}_i^{\text{C}}), F(\mathbf{x}_j^{\text{C}}))}{\mathcal{C}(y_i^{\text{C}}, y_j^{\text{C}})} \tag{2}$$

where $dist(\cdot, \cdot)$ denotes the distance metric used to measure the distance between representations. Both CDA and GSDA emphasize significant label distribution discrepancies, while the GSDA problem places greater emphasis on a minimal difference in confusion features between a minor grade is likely to be confused with a specific major grade.

B. Asymmetric Bi-Classifier Discrepancy Minimization

The objective of a neural network trained with empirical risk minimization is to approximate the underlying conditional probability distribution $P(y|\mathbf{x})$. For class-balanced classifier methods, the aim is to achieve a more favorable approximation of the balanced Bayesian optimal risk, which specifically refers to $\frac{P(y|\mathbf{x})}{P(y)}$.

To mitigate class imbalance, we can make use of an improved class-balanced classifier that introduces class re-balancing schemes (e.g. cost-sensitive learning [29] or logit adjustment [11]). However, these approaches may cause another challenging complication: When the target label distribution closely aligns with the imbalanced source label distribution, an improved classifier is highly likely to yield an increase in per-class mean accuracy but a decrease in overall accuracy due to the majority class with an absolute numerical advantage performing worse.

From the perspective of classifier design, balancing per-class mean accuracy and overall accuracy poses a challenging trade-off. However, by training a more discriminative feature extractor, we can identify the inherent distinguishing features across categories, which can ultimately achieve higher overall accuracy and per-class mean accuracy simultaneously. Equivalently, the discrepancy between the predictions of two different classifiers can be used to evaluate the discriminative power of the feature extractor. A superior feature extractor F^* should predict the same label (ground truth) using two different classifiers based on distinct criteria:

$$\arg\max_{y\in\mathcal{Y}} P(y|(C_{\text{ERM}} \circ F^*(\mathbf{x})) = \arg\max_{y\in\mathcal{Y}} P(y|(C_{\text{BO}} \circ F^*(\mathbf{x})) \tag{3}$$

In other words, when the discriminative ability of the feature extractor is weak, a large number of features passing through the imperfect feature extractor will cluster near the decision boundaries of adjacent categories. The classifiers based on different criteria will come up with wildly different predictions.

Our method is guided by this core intuition to utilize the discrepancy between the predictions of different classifiers to measure the discrimination of the feature extractor (especially the discrimination for adjacent grades with similar features) and force the feature extractor to find the essential features between the grades so that the grading results of the two classifiers are not different. At the instance level, for instances that are highly similar to the source domain, the classification results are not affected by the type of classifier. However, for target instances that are far from the source support or near the original decision boundary, prediction discrepancies will likely arise between the two classifiers.

Using two different criteria, we employ an asymmetric bi-classifier structure to come up with two different decision boundaries. This structural modification of the framework allows for the use of two boundary datum lines instead of one conventional decision boundary, improving the accurate localization of points of confusion near the minority grade. An example of these two boundaries is shown in Fig. 2.

One classifier, C_{ERM}, is based on the general empirical risk minimization strategy, where optimizing the cross-entropy loss L_{CE} is a practical approach to achieving ERM:

$$\min_{\theta_{c_{\text{ERM}}}} \mathcal{L}_{\text{ERM}}(\mathbf{x}, y) = \min_{\theta_{c_{\text{ERM}}}} \mathbb{E}_{\mathbf{x}\sim\mathcal{X}_s} L_{CE}(C_{\text{ERM}} \circ F(\mathbf{x}), y) \tag{4}$$

where θ_{ERM} is the corresponding parameters of C_{ERM}.

The other one is based on the Bayes-optimal balanced risk minimization prediction strategy. The ERM strategy's softmax $\frac{e^{y^i_{\text{out}}}}{\sum_j e^{y^j_{\text{out}}}}$ optimizes $P(y|\mathbf{x})$. To remove label distribution frequency effects, the Bayesian-optimal balanced risk optimization objective should be $\frac{P(y|\mathbf{x})}{P(y)}$. That is, the exponential term $e^{y^i_{\text{out}}}$ in the original formula becomes $e^{y^i_{\text{out}}}/\boldsymbol{\pi}_s^i$, which can be achieved by applying logits adjustment to the outputs, $y_{\text{out}} - \log \boldsymbol{\pi}_s$. The prior offset adjustment of the grade is directly enforced at logits in combination with the known source domain grade label distribution:

$$\min_{\theta_{c_{\text{BO}}}} \mathcal{L}_{\text{Bay}}(\mathbf{x}, y) = \min_{\theta_{c_{\text{BO}}}} \mathbb{E}_{\mathbf{x}\sim\mathcal{X}_s} L_{CE}[(C_{\text{BO}} \circ F(\mathbf{x}) - \log \boldsymbol{\pi}_s), y] \tag{5}$$

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where θ_{BO} is the corresponding network parameters of C_{BO}.

Inspired by the dual classifier adversarial idea [13], [30] and the aforementioned intuition, we fix the classifiers and optimize the feature extractor to generate discriminative features consistent across both classifiers (Fig. 3, Step 2). We use L_1-distance to measure the output discrepancy between C_{ERM} and C_{BO} on $\mathbf{x}_t \in \mathcal{X}_t$:

$$\min_{\theta_f} \mathcal{L}_{\mathrm{dis}}(\mathbf{x}_t) = \min_{\theta_f} \mathbb{E}_y \left\| [p(y|C_{ERM} \circ F(\mathbf{x}_t)) - p(y|C_{BO} \circ F(\mathbf{x}_t))] \right\|_1. \tag{6}$$

By computing the difference between asymmetric dual classifiers, we adequately locate two types of confusing instances that are far from the source support and near the classification boundaries. The purpose of Asymmetric Bi-Classifier Discrepancy Minimization is to optimize the feature extractor such that the classifier outputs for these samples are not different. The target data can generate distinctive features through a superior feature extractor while obtaining consistent predictions in the asymmetric bi-classifier.

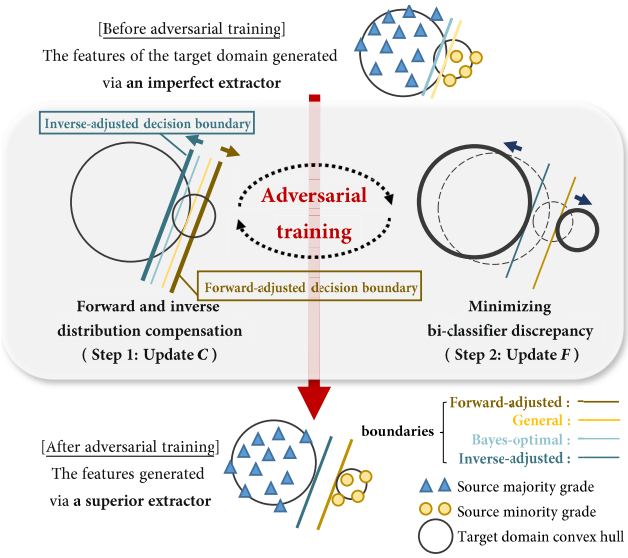


Fig. 3. An overview of the asymmetric bi-classifier adversarial training process. First, identify the confusion instances by the forward and inverse distribution compensation of the dual classifiers. Then, minimize bi-classifier discrepancy in the feature extractor to uncover the essential distinguishing features between the grades.

C. Adversarial Training With Forward and Inverse Distribution Compensation Mechanism

The other primary goal is to maximize the difference between classifiers to identify as many of these easily ambiguous instances as possible. The imbalance of the dataset leads to biased and unreliable pseudo-labels on the target domain. Unlike conventional dual classifier adversarial methods, we no longer use the discrepancy of pseudo-labels on the target data to maximize classifier discrepancy. Instead, we amplify the criteria differences of asymmetric classifiers using accurately labeled source domain data.

First, to estimate whether the source domain is imbalanced, we calculate the L_1-distance $T_s(\boldsymbol{\pi}_s)$ between the source label distribution $\boldsymbol{\pi}_s$ and the balanced label distribution, $T_s(\boldsymbol{\pi}_s) = \left\| [\sum_{k=0}^{V-1} (\boldsymbol{\pi}_s^k - \frac{1}{V}) \right\|_1$ (V for DR grading problem). When the $T_s(\boldsymbol{\pi}_s)$ is below the pre-defined threshold I_T, the maximum classifier discrepancy [13] is used to amplify the classifier difference. When $T_s(\boldsymbol{\pi}_s)$ above the threshold I_T, we introduce the Forward and Inverse Distribution Compensation (FIDC) mechanism.

As shown in Fig. 3 step 1, our objective is to separate the classifiers from each other, as the primary factor contributing to the asymmetric classifier difference is the imbalanced distribution. The extent of skewness will directly impact the positioning of the classification boundaries.

1) Forward Distribution Compensation for C_{ERM}: We perform forward source prior distribution compensation on the original distribution by oversampling the majority grades and undersampling the minority grades. The optimization objective corresponding is $P(y|\mathbf{x}) \cdot P(y^+)^{\mu}$, where $\mu > 0$ is a hyper-parameter to weigh the extent to the distribution compensation. Let $\boldsymbol{\pi}_s^+$ denote the forward distribution compensation, which is obtained by further applying logits adjustment with $\boldsymbol{\pi}_s$. Similarly, we can adjust the output of C_{ERM} to $(y_{\mathrm{out}} + \mu \log \boldsymbol{\pi}_s^+)$:

$$\mathcal{L}_{\mathrm{ERM+}}(\mathbf{x}, y) = \mathbb{E}_{\mathbf{x} \sim \mathcal{X}} - y_i \log(C_{\mathrm{ERM}} \circ F(\mathbf{x}) + \mu \log \boldsymbol{\pi}_s^+) \tag{7}$$

Intuitively, the resulting predictions exhibit a stronger bias towards the majority grades, causing the decision boundaries to shift towards the minority grades, and vice versa.

2) Inverse Distribution Compensation for C_{BO}:

$$\mathcal{L}_{\mathrm{Bay\text{-}}}(\mathbf{x}, y) = \mathbb{E}_{\mathbf{x} \sim \mathcal{X}} - y_i \log(C_{\mathrm{BO}} \circ F(\mathbf{x}) - \log \boldsymbol{\pi} + \mu \log \boldsymbol{\pi}_s^-) \tag{8}$$

$\boldsymbol{\pi}_s^-$ denotes the inverse distribution compensation, which is obtained by applying logits adjustment in the reverse direction with $\boldsymbol{\pi}_s$. Inverse-adjusted predictions are more biased toward the minority grade than balanced Bayes-optimal predictions.

On the one hand, asymmetric bi-classifier adversarial training does not require maximizing the discrepancy to generate the differentiation between the two classifiers. The differentiation arises from adjusting the logits based on the imbalanced source domain, resulting in a natural "river" between grades. This enhances the clarity and separability of features in the latent space across different grades after training.

On the other hand, the FIDC mechanism potentially aids in the identification of the targets that exhibit greater ambiguity in proximity to the decision boundary and makes it easier for the model to adapt to a wider range of label distribution shifts centered on the source domain prior label distribution $\boldsymbol{\pi}_s$.

D. Overall Training

To sum up, the overall adversarial training process for each epoch with augmented data (i.e. ABiD‡) can be summarized as follows (refer to Fig. 3):

First, we define the sum of cross-entropy loss obtained by the asymmetric bi-classifier, which involves feeding the

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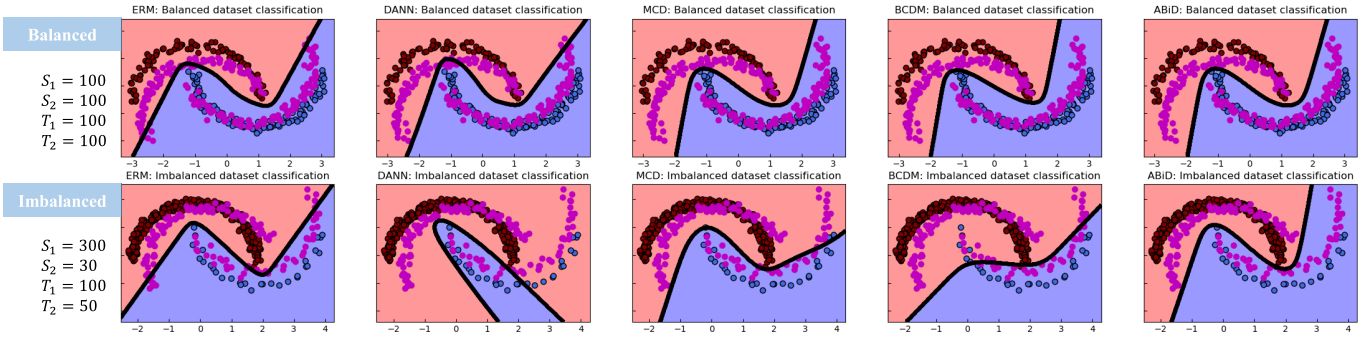


Fig. 4. Visualizations of comparisons of decision boundaries on balanced and imbalanced inter-twining moons datasets. The purple dots represent the target domain data, and the red and blue dots represent the source data points belonging to classes 1 and 2, respectively.

data through the shared feature extractor and then separately applying the features to the two classifiers, as:

$$\mathcal{L}_s(\mathbf{x}_s, y) = L_{\text{CE}}(C_{ERM} \circ F(\mathbf{x}_s), y) + L_{\text{CE}}(C_{BO} \circ F(\mathbf{x}_s), y) \tag{9}$$

Step 1: Update C_{ERM}, C_{BO} through the FIDC mechanism to increase the disparity between their predictions. At the same time, the feature extractor parameters θ_f are remain fixed:

$$\min_{\theta_{c_{\text{ERM}}}, \theta_{c_{\text{BO}}}} \frac{1}{4} \left[\mathcal{L}_{\text{ERM+}}(\{\mathbf{x}_s, \widetilde{\mathbf{x}}_s\}, y) + \mathcal{L}_{\text{BO-}}(\{\mathbf{x}_s, \widetilde{\mathbf{x}}_s\}, y) \right] \tag{10}$$

Step 2: After leveraging forward and inverse distribution compensation to amplify the disparity between the classifiers, we update the feature extractor parameters θ_f by minimizing the disparity in classifier predictions on the target domain $\mathbf{x}_t \in \mathcal{X}_t$ under the source supervision:

$$\min_{\theta_f} \frac{1}{2} \mathcal{L}_s(\{\mathbf{x}_s, \widetilde{\mathbf{x}}_s\}, y) + \mathcal{L}_{\text{dis}}(\mathbf{x}_t) \tag{11}$$

Step 3: Finally, the source risk is re-minimized to mitigate the cumulative impact of the compensation mechanism on the classifiers:

$$\min_{\theta_f, \theta_{c_{\text{ERM}}}, \theta_{c_{\text{BO}}}} \frac{1}{2} \mathcal{L}_s(\{\mathbf{x}_s, \widetilde{\mathbf{x}}_s\}, y) \tag{12}$$

The main procedures are summarized in Algorithm 1. Correspondingly, the training procedure for the asymmetric bi-classifier without augmented data (i.e. ABiD) is to substitute $\widetilde{\mathbf{x}}_s$ with $\mathbf{x}_s$, *e.g.*, use $\mathcal{L}_s(\mathbf{x}_s)$ instead of $[\mathcal{L}_s(\{\mathbf{x}_s, \widetilde{\mathbf{x}}_s\}, y)]/2$.

Algorithm 1 ABiD's Main Learning Algorithm

Input: batch size N, source domain $\{\mathbf{x}_s, y\}$ and target domain $\{\mathbf{x}_t\}$, learning rate γ_1, γ_2, and hyper-parameter μ.
Output: Optimal parameters θ_f^*, $\theta_{c_{\text{ERM}}}^*$, $\theta_{c_{\text{BO}}}^*$

```
Initialize parameters θ_f, θ_cERM, θ_cBO.
while epoch_count<epoch do
  for i = 1 to C_times do
    Calculate the FIDC in Eq.(10) to update θ_cERM, θ_cBO:
      θ_c1 ← θ_c1 − γ2 ∂(L_ERM+ + L_BO-)/∂θ_c1
      θ_c2 ← θ_c2 − γ2 ∂(L_ERM+ + L_BO-)/∂θ_c2
  end for
  for j = 1 to F_times do
    Minimize discrepancy in Eq.(11) to update θ_f.
      θ_f ← θ_f − γ1 ∂(L_s + L_dis)/∂θ_f
  end for
  Minimize the source risk in Eq.(12):
      θ_f ← θ_f − γ1 ∂L_s/∂θ_f
      θ_c1 ← θ_c1 − γ2 ∂L_s/∂θ_c1,  θ_c2 ← θ_c2 − γ2 ∂L_s/∂θ_c2
end while
return θ_f, θ_cERM, θ_cBO
```

IV. Experimental Evaluation

In this section, we evaluate the ABiD method with 21 DA tasks on a toy dataset and three real-world datasets. The code and the partition of the DDR dataset can be found at *https://github.com/FByyyyuan/ABiD*. We conduct extensive experiments to address the following research questions.

- **RQ1:** How does our method perform in the presence of covariate shift, grade skew, and small lesion challenges? (Table II, III, IV and Fig. 4)
- **RQ2:** Does our method improve performance across all grades under grade skew conditions? (Table V)
- **RQ3:** How effectively is our method detecting small lesions? (Fig. 5)
- **RQ4:** How effective are the individual components of our method? (Table VI and VII)
- **RQ5:** Is our method composable and versatile? (Table VIII and IX)

A. Experiments on Toy Datasets

We devised a toy example that effectively visually illustrates the reasons behind the superior performance of the ABiD method and elucidates how this method facilitates more accurate classification for specific instances. We evaluated our ABiD method against prior dual classifier adversarial arts, namely ERM [31], DANN [12], MCD [13] and BCDM [15], in terms of performance on two distinct domain-shifted binary classification datasets: a balanced inter-twining moons dataset and an imbalanced inter-twining moons dataset. In the toy dataset experiments, each method's feature extractor is a single fully connected layer that projects the coordinate inputs into a 1×15 feature vector, then processed by the respective classifier systems.

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TABLE I
THE DETAILED INFORMATION (SAMPLE SIZE ^{PROPORTION}) OF THE REAL-WORLD DATASETS

	DR0	DR1	DR2	DR3	DR4	Sum	$T_s(\pi)$
IDRiD-S	134 $^{0.32}$	20 $^{0.05}$	136 $^{0.33}$	74 $^{0.18}$	49 $^{0.12}$	413	0.50
IDRiD-T	34 $^{0.33}$	5 $^{0.05}$	32 $^{0.31}$	19 $^{0.18}$	13 $^{0.13}$	103	0.48
DDR-1$_{(2048\times1536)}$	487 $^{0.34}$	94 $^{0.07}$	735 $^{0.51}$	26 $^{0.02}$	100 $^{0.07}$	1442	0.89
DDR-2$_{(2736\times1824)}$	1435 $^{0.67}$	92 $^{0.04}$	507 $^{0.24}$	34 $^{0.02}$	61 $^{0.03}$	2129	1.02
DDR-3$_{(2960\times2935)}$	290 $^{0.57}$	33 $^{0.06}$	146 $^{0.29}$	8 $^{0.02}$	31 $^{0.06}$	508	0.92
Aier-Site A	819 $^{0.59}$	122 $^{0.09}$	127 $^{0.09}$	67 $^{0.05}$	250 $^{0.18}$	1385	0.78
Aier-Site B	179 $^{0.38}$	27 $^{0.06}$	105 $^{0.22}$	85 $^{0.18}$	71 $^{0.15}$	467	0.41
Aier-Site C	142 $^{0.25}$	33 $^{0.06}$	250 $^{0.44}$	48 $^{0.09}$	90 $^{0.16}$	563	0.58
Aier-Site D	1251 $^{0.56}$	250 $^{0.11}$	249 $^{0.11}$	250 $^{0.11}$	249 $^{0.11}$	2249	0.72

The visualization results are shown in Fig. 4. The target instances (purple points) are generated by rotating the instances of the source domain.

For the balanced dataset, in both the source and target domains, each class has 100 instances.

For the imbalanced dataset, we designed the dataset based on the specific characteristics of GSDA: the source domain consists of two classes with sample sizes of 300 and 30, while the target domain comprises two classes with sample sizes of 100 and 50, respectively. The balanced and imbalanced datasets used the same model configuration.

Evidently, for the balanced dataset, all DA methods demonstrated improvements over the ERM. Moreover, employing the two classifiers adversarial approaches such as MCD, BCDM and our method led to 100% classification accuracy.

However, in the imbalanced dataset, all the compared DA methods exhibited a noteworthy decline in performance compared to the balanced dataset, with a tendency for data to be misclassified into the majority class. MCD misclassified all instances in the upper-right corner, showing clear bias towards Class 1 due to the unfair premise of pseudo-labels amplifying classifier differences. BCDM proposes to use CDD instead of L_1-distance in MCD to calculate the disparity between the two classifiers. This modification increases distinguishability but, at the same time, aggravates discrimination against minority classes when dealing with class imbalance, as the minority classes are more likely to be abandoned. However, even though this imbalanced dataset is challenging, our method yielded a perfect decision boundary and was still capable of achieving 100% classification accuracy.

B. Real-World Datasets

We conducted comprehensive experiments to compare our method against the state-of-the-art methods on two public datasets *IDRiD, DDR* and a private dataset *Aier*. The sample size per grade in each domain is shown in Table I.

1) Indian Diabetic Retinopathy Image Dataset, IDRiD: For the smaller Indian Diabetic Retinopathy Image Dataset (IDRiD) [32], we use the original training and test set split, where the training set serves as the source domain and the test set as the target domain. This dataset has no obvious difference in acquisition equipment, but due to the complexity of DR grading task itself, the model trained on the source domain still cannot be perfectly classified in the target domain.

2) Dataset for Diabetic Retinopathy Classification, DDR: We give a domain partition for DDR, which is convenient for other studies to analyze the domain shift in practice. We divide the original Dataset for Diabetic Retinopathy Classification (DDR) [33] into different domains. For the public datasets, there is currently no large-scale DR dataset explicitly designed for domain adaptation task. Therefore, to obtain a clear domain partition, we give a domain partition for DDR, which is convenient for other studies to analyze the domain shift in practice. Fundus images can be captured using different devices, each with its own resolutions and mask configurations. The specific characteristics of the fundus images depend on the device used for capturing them. Therefore, we selected a set of images that had the same resolution and invalid region mask as a domain, as these images were very likely to have been collected using the same device. We selected three cross-device disparate domains that have a large number of samples, with resolutions of (2048×1536), (2736×1824), and (2960×2935), respectively.

3) Private Dataset Provided by Aier EYE Hospital, Aier: The private dataset provided by Aier EYE Hospital (Aier) is composed of sub-datasets from four geographically distant medical institutions, each naturally representing a distinct domain. These domain variations encompass diverse imaging protocols and devices, as well as patient populations. In the context of fundus imaging, different populations may have distinct characteristics, such as variations in demographics, genetic factors, lifestyle, and prevalence of specific eye conditions. Such situations are common in practical deployment, where performance degradation occurs when the trained model is applied across medical institutions. To thoroughly evaluate the effectiveness of our method in addressing domain shifts in realistic cross-center scenarios, we employed the Aier dataset obtained from multiple real healthcare centers.

C. Baselines

We compared our method with the state-of-the-art methods. Specifically, the comparative methods encompass several research areas: **Class Re-balancing methods:** Logit Adjustment (LA) [11], Self-supervised aggregation of diverse experts (SADE) [34]. **Unsupervised Domain adversarial adaptation methods:** Domain-Adversarial Neural Network (DANN) [12], Maximum Classifier Discrepancy (MCD) [13], Bi-Classifier Determinacy Maximization (BCDM) [15]. **Joint distribution adaptation (class-imbalanced domain adaptation) methods:** Joint Adaptation Network (JAN) [20], CO-ALignment (COAL) [16], Selective Entropy Optimization (SENTRY) [21], Pairwise Adversarial Training (PAT) [22], Adversarial training on Penultimate Activations (APA) [23], Backprop Induced Feature Weighting for Adversarial Domain Adaptation (BIWAA) [24], Gradual Source Domain Expansion (GSDE) [25]. **Other methods:** We also compared ERM baseline [31], [35] and a classical unsupervised domain adaptation method Batch Nuclear-norm Maximization (BNM) [36].

Except for methods SENTRY and PAT, image augmentation is a non-essential module. SENTRY necessitates the integration of the data augmentation strategy, with RandAugment [37] serving as the employed augmentation method. PAT proposes

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a new data augmentation plug-in method, in which domain adaptive loss can be replaced by many DA methods. We use default MDD + Implicit Alignment [38], [39] loss.

D. Configuration

1) Experimental Settings: Our framework was implemented in Python with PyTorch. To ensure fairness and consistency, we adopted pretrained ConvNeXt-Tiny [35] backbone across all compared methods on all datasets. We employed the batch size of 32 and conducted training for a duration of 70 epochs across our method and other methods. For all experiments, we set the learning rate for the feature extractor and classifier layers to 0.0001 and 0.001, respectively. All parameters were updated via the Adam optimizer with weight decay of 0.0001 and 0.0005 for the feature extractor and two asymmetric classifiers, respectively. We set μ to 0.5, and set both F_{times} and C_{times} to 2. All results were reported with average score over 3 trials. I_T was preset to 0.4 and all domains on the three datasets were judged as imbalanced domains. All the experiments were implemented on Nvidia RTX 3090 platform.

2) Evaluation Metrics: We used several evaluation metrics to measure the performance of the model: Overall Accuracy (Acc.), Per-grade Mean Accuracy (PAcc.) and Quadratic-weighted Kappa score (κ) [4], which are often used to evaluate the effect of grading medical images [40], [41]. κ is calculated as:

$$\kappa = 1 - \frac{\sum_{i=1}^{V}\sum_{j=1}^{V} O(i,j)W(i,j)}{\sum_{i=1}^{V}\sum_{j=1}^{V} E(i,j)W(i,j)} \tag{13}$$

where V is the number of grades, $O(i,j)$ is the histogram matrix, $E(i,j)$ is the expected matrix [4] and $W(i,j)=\frac{(i-j)^2}{(V-1)^2}$ is the weight matrix penalizing prediction errors.

3) Image Augmentation: To fairly compare with the comparison method with and without image augmentation, we trained with and without augmented data (namely ABiD‡ and ABiD), respectively. We employed various augment techniques commonly used in retinal fundus image processing sequentially, including random rotation (degrees range from -35 to 35), random hue alteration (hue factor is chosen uniformly from 0 to 0.25), random conversion to grayscale, random horizontal flipping, image smoothing using Gaussian blur, and random resized cropping. For the Gaussian blur transformation module, we used the Gaussian kernel of 3×3 size, and the standard deviation was selected uniformly at random in the range of (1, 2). Before feeding into the backbone network, we resized all instances into the size of 224×224.

E. Result and Discussion

Table II (sorted by method type) shows the experimental results on IDRiD. We observe that grading results of the two classifiers tend to be the same after convergence. The grading results of C_{ERM} is reported. Table III and Table IV (sorted by Acc. value) present the experimental results of 6 cross-domain adaptation tasks on the DDR dataset and 12 cross-domain adaptation tasks on the Aier dataset, respectively.

TABLE II
RESULTS (AVERAGE$^{\pm S.D.}$) ON IDRID WITH CONVNEXT-TINY BACKBONE. DOUBLE DAGGER (‡) DENOTES THE METHODS USING AUGMENTED DATA. THE BEST RESULTS W/ AUGMENTATION ANNOTATED IN **RED**, BEST RESULTS W/O AUGMENTATION ANNOTATED IN **BLUE**

Method	Acc.	Per-grade Acc.	Q.W.kappa
LA, 2021 [11]	$55.02^{\pm 1.29}$	$49.17^{\pm 1.86}$	$68.97^{\pm 3.92}$
SADE, 2022 [34]	$53.40^{\pm 3.88}$	$49.27^{\pm 1.05}$	$73.15^{\pm 1.16}$
DANN, 2016 [12]	$56.31^{\pm 2.91}$	$41.73^{\pm 0.81}$	$64.76^{\pm 7.66}$
MCD, 2018 [13]	$55.66^{\pm 2.27}$	$42.39^{\pm 0.84}$	$68.22^{\pm 5.58}$
BCDM, 2021 [15]	$57.61^{\pm 3.24}$	$44.34^{\pm 3.83}$	$67.98^{\pm 1.09}$
JAN, 2017 [20]	$56.63^{\pm 2.59}$	$48.24^{\pm 0.82}$	$67.43^{\pm 0.38}$
COAL, 2020 [16]	$59.43^{\pm 3.12}$	$45.34^{\pm 3.58}$	$66.74^{\pm 1.74}$
SENTRY‡, 2021 [21]	$54.05^{\pm 3.12}$	$49.22^{\pm 3.58}$	$66.35^{\pm 1.74}$
PAT‡, 2022 [22]	$59.55^{\pm 1.62}$	$50.36^{\pm 2.64}$	$71.65^{\pm 1.02}$
APAn, 2023 [23]	$57.93^{\pm 2.97}$	$46.47^{\pm 1.75}$	$70.71^{\pm 1.00}$
BIWAA-I, 2023 [24]	$60.52^{\pm 4.59}$	$48.63^{\pm 4.29}$	$71.96^{\pm 2.20}$
GSDE, 2024 [25]	$59.55^{\pm 2.24}$	$46.15^{\pm 0.31}$	$73.14^{\pm 1.02}$
ERM, 1999 [31], [35]	$57.28^{\pm 3.88}$	$42.31^{\pm 2.62}$	$63.92^{\pm 1.76}$
BNM, 2020 [36]	$51.78^{\pm 2.27}$	$38.65^{\pm 1.12}$	$65.54^{\pm 1.08}$
ABiD (Ours)	**$64.72^{\pm 2.59}$**	**$49.29^{\pm 2.53}$**	**$73.92^{\pm 2.23}$**
ABiD‡ (Ours)	**$65.70^{\pm 1.29}$**	**$52.65^{\pm 2.83}$**	**$76.80^{\pm 2.48}$**

1) Comparison With ERM: Compared with the ERM, our method exhibits superior performance across all datasets on average. We utilize the distinctions generated by asymmetric bi-classifiers to evaluate the extractor, fostering intrinsic motivation for discovering essential differences and mitigating the impact of grade-skewed domain shift. Specifically, the most significant performance improvements attained with our ABiD (ABiD‡) method were 7.44% (8.42%), 2.45% (5.05%), 3.67% (4.52%) in term of overall accuracy (Acc.) and 10.00% (12.88%), 6.19% (9.25%), 3.06% (3.61%) in term of quadratic-weighted kappa (κ) for each of the three datasets, respectively. Meanwhile, the average performance of many compared methods, even those that take label distribution shift into account, is worse than ERM.

2) Comparison With the State-of-the-Art: The ABiD‡ method, with augmented data, surpasses all state-of-the-art (SOTA) methods and achieves the highest performance across evaluated metrics and datasets. The ABiD method, without augmented data, yields suboptimal model performance.

We can observe that many of the current DA methods cannot effectively solve the problem of grade-skewed domain adaptation in DR grading, even the methods considering label shift. For example, the SENTRY comprises a committee of multiple different random augments, and the selective entropy optimization is carried out according to the consistency of prediction. In essence, the SENTRY still relies on the pseudo-labels of the target domain given by the committee of multiple augments. Since features are more likely to diverge, this method has achieved good results in the CDA problem. However, in the GSDA problem, because small lesions are not easy to find, the features between adjacent grades are more similar, and there are also obvious quantitative differences between grades, which will make the whole committee still biased towards the major grades in adjacent grades. Some representative DA methods like MCD and BCDM show very low PAcc. results,

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TABLE III
RESULTS ON THE DDR DATASET WITH CONVNEXT-TINY BACKBONE (SOURCE DOMAIN→ TARGET DOMAIN)

Method	Domain 1→2			Domain 1→3			Domain 2→1			Domain 2→3			Domain 3→1			Domain 3→2			**AVG**		
	Acc.	PAcc	κ	Acc.	PAcc	κ	Acc.	PAcc	κ	Acc.	PAcc	κ	Acc.	PAcc	κ	Acc.	PAcc	κ	Acc.↑	PAcc	κ
SADE	78.6	52.9	73.1	49.3	54.1	64.3	54.1	45.2	54.2	55.8	52.4	62.3	57.2	43.1	58.0	75.2	52.9	70.2	61.7	50.1	63.7
LA	80.3	49.5	72.5	61.9	53.6	62.5	58.1	48.1	52.6	55.6	53.0	64.3	52.7	44.1	56.9	71.4	**54.4**	70.6	63.3	50.4	63.2
JAN	62.1	47.1	45.5	64.2	50.8	58.5	56.5	39.2	42.8	69.2	46.2	66.2	61.9	39.5	53.4	80.3	49.9	**75.8**	65.7	45.5	57.0
BNM	79.8	47.8	69.9	64.5	49.5	60.0	55.1	39.3	46.7	63.5	50.4	54.8	60.5	40.5	55.0	79.3	46.1	67.8	67.1	45.6	59.0
APAn	77.0	40.9	60.3	69.5	47.5	60.7	57.1	36.7	47.7	65.2	45.3	66.5	63.7	42.5	56.7	73.5	46.0	65.8	67.7	43.2	59.6
DANN	79.4	45.3	66.8	62.9	38.4	57.7	61.3	43.3	52.4	69.2	33.6	55.5	64.5	35.6	50.6	77.9	39.1	65.5	69.2	39.2	58.1
BCDM	80.4	42.7	67.8	71.8	46.8	68.8	57.9	46.8	47.8	68.4	39.4	56.2	65.1	38.0	55.8	78.8	38.3	63.9	70.4	42.0	60.1
ERM	80.7	44.0	69.8	69.3	42.4	66.1	61.1	39.5	49.5	67.1	47.8	58.1	66.5	41.8	54.2	78.4	43.2	66.1	70.5	43.1	60.6
GSDE	80.5	51.1	72.4	67.4	49.7	63.0	59.7	**51.1**	56.9	71.7	**55.0**	68.8	67.1	45.0	60.0	78.8	45.7	69.6	70.9	49.6	65.1
BIWAA-I	80.6	47.7	70.3	69.6	48.1	64.5	61.3	45.2	**54.9**	70.9	51.9	68.2	64.1	43.3	58.9	**81.0**	47.0	75.1	71.3	47.2	65.3
COAL	80.6	40.9	69.1	71.5	42.2	66.2	**63.1**	33.2	49.2	68.2	29.1	57.1	**68.8**	40.4	57.2	77.1	31.7	58.7	71.5	36.2	59.6
MCD	81.3	44.8	69.2	68.0	47.3	59.5	62.0	37.2	53.0	73.0	43.7	61.2	66.4	39.7	57.0	79.5	41.2	58.4	71.7	42.3	59.7
ABiD	**82.0**	**51.6**	**73.2**	**74.1**	**54.6**	**73.8**	60.2	43.7	51.3	**73.2**	53.9	**71.0**	68.7	**49.6**	**62.8**	79.6	50.0	68.9	**73.0**	**50.5**	**66.8**
PAT‡	79.6	55.6	**75.7**	63.1	56.4	69.3	54.0	**52.6**	60.1	59.6	54.7	65.8	48.1	49.4	59.0	61.2	52.5	64.7	60.9	53.5	65.8
SENTRY‡	73.4	48.8	66.7	64.6	44.5	66.7	50.0	44.2	50.1	60.6	51.2	63.5	57.6	42.2	57.3	76.6	49.3	66.7	63.8	46.7	61.8
ABiD‡	**83.0**	**56.9**	75.0	**73.6**	**60.0**	**73.2**	**69.2**	48.6	**60.9**	**77.4**	**56.6**	**75.6**	**69.7**	**51.1**	**63.7**	**80.5**	**54.0**	**71.0**	**75.6**	**54.5**	**69.9**

TABLE IV
RESULTS ON THE AIER DATASET WITH CONVNEXT-TINY BACKBONE (SOURCE DOMAIN→TARGET DOMAIN)

Method	A→B			A→C			A→D			B→A			B→C			B→D			C→A			C→B			C→D			D→A			D→B			D→C			**AVG**		
	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.	PAcc.	κ	Acc.↑	PAcc.	κ
JAN	57.3	44.5	68.2	48.2	42.0	54.7	74.5	55.0	89.9	61.0	48.0	72.6	62.9	52.0	74.0	71.5	54.0	84.9	57.4	45.3	70.5	63.7	52.3	84.2	62.4	52.5	79.3	76.5	57.9	89.7	63.2	54.2	75.1	51.5	43.3	60.8	62.5	50.1	75.3
LA	38.7	38.2	72.8	**66.5**	45.3	70.5	63.3	53.5	85.4	68.6	52.0	82.4	63.9	56.2	75.9	71.6	59.6	88.5	69.0	**56.5**	85.5	73.9	62.9	92.4	70.5	61.4	90.7	73.9	57.1	89.5	72.4	70.5	90.9	63.9	**63.0**	83.2	66.4	56.4	84.0
BNM	58.0	51.6	76.5	55.2	**48.8**	67.7	60.2	41.3	68.4	**72.6**	**53.1**	**86.6**	60.1	**57.2**	76.9	71.5	53.3	86.5	71.7	46.5	84.6	73.5	60.4	90.9	68.1	50.4	83.8	76.8	**59.1**	89.8	72.4	69.7	89.5	59.8	57.9	81.8	66.7	54.1	81.9
SADE	49.1	42.7	78.6	64.4	46.2	**72.6**	71.5	55.8	90.2	67.4	50.8	82.1	58.8	55.8	75.5	72.2	59.1	88.9	66.4	55.5	83.2	72.0	64.1	92.3	72.4	61.6	90.5	73.2	56.5	89.9	73.7	69.1	91.7	69.2	63.7	84.2	67.5	56.8	85.0
DANN	65.5	49.0	80.1	59.1	42.9	60.7	69.4	45.1	85.3	69.5	48.6	80.7	67.6	49.1	71.9	72.7	53.2	87.1	67.9	45.6	80.1	68.8	52.5	87.0	68.0	53.4	84.8	**77.5**	57.4	90.2	71.9	66.3	90.3	54.9	47.2	72.1	67.8	50.9	80.9
BCDM	48.3	41.4	75.2	65.2	46.1	67.1	71.7	52.1	89.2	71.0	49.6	82.4	68.3	51.4	73.7	74.8	56.4	88.5	69.0	46.5	80.2	74.3	60.3	92.2	74.7	58.3	91.0	70.5	58.0	88.0	71.9	66.9	90.3	64.3	55.4	79.6	68.7	53.5	83.1
ERM	47.6	43.2	79.0	65.6	45.9	65.6	71.1	53.4	89.6	69.8	50.0	80.5	69.6	53.1	74.6	73.6	57.4	87.4	69.6	49.3	79.8	75.0	61.3	90.7	74.3	59.7	90.4	74.4	56.7	89.8	72.6	63.6	91.8	63.5	61.6	82.2	68.9	54.6	83.4
COAL	60.7	46.2	78.4	56.7	35.7	56.2	71.9	49.7	88.2	71.1	47.7	82.7	69.3	48.3	72.4	73.0	53.4	88.2	71.6	48.8	82.8	75.8	61.5	92.7	76.1	59.0	91.8	73.8	52.0	88.6	71.2	68.1	90.7	61.3	58.2	79.2	69.3	52.4	82.6
BIWAA-I	64.3	51.5	82.9	62.3	45.6	70.5	71.9	53.3	89.1	70.0	51.4	83.5	65.7	56.4	80.7	72.8	59.8	89.2	70.1	52.8	82.8	77.2	**65.6**	93.4	74.6	58.3	90.7	73.0	55.8	89.8	74.9	**71.8**	91.9	62.5	60.7	82.4	69.9	56.9	85.6
GSDE	64.5	52.5	84.5	63.3	51.3	70.9	70.5	54.3	90.9	69.5	50.8	81.7	64.4	54.9	76.6	74.1	60.0	90.0	67.0	51.3	80.9	**77.0**	63.7	**93.5**	75.3	61.6	91.5	73.8	56.9	90.1	74.6	71.6	91.0	61.2	59.8	83.0	69.6	57.3	85.4
MCD	61.2	48.8	79.9	64.0	44.4	63.1	73.1	52.0	89.8	71.3	51.5	83.1	67.1	53.0	74.5	74.8	57.7	88.2	71.0	49.0	81.5	74.5	59.1	91.5	76.3	58.8	92.4	75.1	56.9	90.7	**75.9**	71.3	91.9	63.9	58.9	82.0	70.7	55.1	84.1
ABiD	**65.5**	**52.5**	**84.6**	65.1	46.2	68.9	**74.6**	**55.1**	**91.1**	**72.6**	**53.1**	84.5	**71.5**	56.3	**78.3**	**75.3**	**60.4**	**90.2**	**72.9**	55.9	**85.6**	74.5	62.9	92.0	**77.4**	**61.8**	**92.7**	76.5	58.1	**91.4**	75.3	69.4	**92.7**	**69.7**	61.6	**85.9**	**72.6**	**57.8**	**86.5**
SENTRY‡	59.3	50.0	81.1	57.8	**50.8**	68.5	73.1	53.9	90.9	68.5	50.8	83.1	60.6	53.9	77.5	66.2	60.0	88.1	65.6	46.0	83.0	68.0	59.4	90.1	50.7	48.4	81.8	73.7	55.1	89.6	56.4	54.8	88.5	67.4	58.3	84.6	63.9	53.8	83.9
PAT‡	67.8	56.3	86.3	64.2	47.3	70.8	73.6	54.6	90.0	68.2	50.9	82.3	64.6	48.3	**80.6**	74.0	**60.3**	90.3	71.9	54.5	83.4	**78.9**	62.3	**94.2**	77.9	63.2	93.0	74.9	55.7	89.6	76.2	70.2	91.3	66.3	**62.9**	84.9	71.5	57.2	86.4
ABiD‡	**70.9**	**58.0**	**88.3**	**67.6**	46.9	**70.9**	**74.9**	**55.2**	**91.2**	**70.5**	**63.3**	**82.5**	**71.8**	**55.2**	78.8	**75.3**	59.6	**90.7**	**72.9**	**55.4**	**85.9**	76.2	**63.5**	93.1	**78.2**	**63.3**	**93.2**	**76.8**	**58.3**	**91.3**	**77.9**	**73.6**	**93.1**	**68.1**	62.5	**85.6**	**73.4**	**58.7**	**87.1**

TABLE V
RESULTS ON EACH GRADE (AIER: D→B)

	DR0	DR1	DR2	DR3	DR4	Acc.	PAcc.
ERM	97.0	41.9	63.1	38.0	77.6	63.6	72.6
LA	93.7$_{\downarrow 3.3}$	79.6$_{\uparrow 37.7}$	47.1$_{\downarrow 16.1}$	41.5$_{\uparrow 3.4}$	90.4$_{\uparrow 12.8}$	72.4	70.5
ABiD‡	98.3$_{\uparrow 1.3}$	73.1$_{\uparrow 31.2}$	67.0$_{\uparrow 3.8}$	42.7$_{\uparrow 4.6}$	86.8$_{\uparrow 9.1}$	77.9	73.6

indicating that the bias towards the majority class observed in the toy dataset also exists in real-world datasets.

Moreover, we also find that several of comparison methods considering label distribution shift perform better under the per-grade mean accuracy evaluation metric, which is close to the evaluation metric in the related literature. Comparably, class re-balancing methods (LA and SADE) have demonstrated favorable results in terms of per-grade mean accuracy. However, when applied to target domains with comparable skewed distributions, the overall accuracy is not competitive.

As shown in Table V, the LA method enhances performance for minority classes but exhibits a decrease in performance for majority classes (DR0, DR2), suggesting a potential bias towards the minority classes and a lack of essential feature learning. In contrast, our method demonstrates substantial performance improvements across all classes, as evidenced by the increased performance in all grades.

F. Ablation Studies and Further Studies

We further conducted ablation experiments on the innovation components of our method and parameter sensitivity analysis in this section.

TABLE VI
ABLATION STUDY ON EACH COMPONENT WITH CONVNEXT-TINY

		Accuracy	PAcc.	Q.W.kappa
1	Baseline	$57.28^{\pm 3.88}$	$42.31^{\pm 2.62}$	$63.92^{\pm 1.76}$
2	ABiD Framework	$58.58^{\pm 2.59}$	$45.91^{\pm 2.40}$	$66.99^{\pm 3.37}$
3	ABiD Framework+ min Dis(T)	$60.19^{\pm 1.94}$	$45.59^{\pm 2.91}$	$68.80^{\pm 4.20}$
4	ABiD Framework+ FIDC (ABiD)	$64.72^{\pm 2.59}$	$49.29^{\pm 2.53}$	$73.92^{\pm 2.23}$
5	ABiD +Aug (ABiD‡)	$65.70^{\pm 1.29}$	$52.65^{\pm 2.83}$	$76.80^{\pm 2.48}$
- Ablation of F: w/ different discrepancy minimization mechanisms				
6	ABiD $F_{\text{times}} \to 0$	$58.58^{\pm 2.59}$	$45.91^{\pm 2.40}$	$66.99^{\pm 3.37}$
7	ABiD $F_{\text{times}} \to 1$	$62.14^{\pm 2.91}$	$48.35^{\pm 1.95}$	$73.01^{\pm 1.35}$
- Ablation of C: w/ different classifier discrepancy amplification strategy				
8	ABiD $C_{\text{times}} \to 0$	$59.87^{\pm 3.24}$	$44.64^{\pm 5.18}$	$64.60^{\pm 1.54}$
9	ABiD $C_{\text{times}} \to 1$	$59.87^{\pm 1.29}$	$46.05^{\pm 2.83}$	$71.62^{\pm 2.48}$
10	ABiD $C_{\text{Max Dis(T)}}$	$57.93^{\pm 2.27}$	$46.21^{\pm 3.28}$	$67.67^{\pm 2.30}$

1) Ablation Studies: We conducted ablation experiments in Table VI. The **Baseline** uses ERM for risk minimization. The last two rows show our methods, **ABiD and ABiD‡**, with and without augmentation. The **ABiD Framework** extends the **Baseline** by implementing our model structure without adversarial training, showing improvements but still lagging behind ABiD. The **ABiD Framework+min Dis(T)** uses an adversarial method for Maximum Classifier Discrepancy, which decreases performance in some metrics, indicating that traditional dual classifier methods may not help with class imbalance. In contrast, **ABiD Framework+FIDC** significantly improves all metrics. Moreover, we further conducted more detailed ablation experiments on these several modules:

a) Asymmetric bi-classifier discrepancy minimization on F: As shown in Table VI (line 6-7), we conducted an ablation

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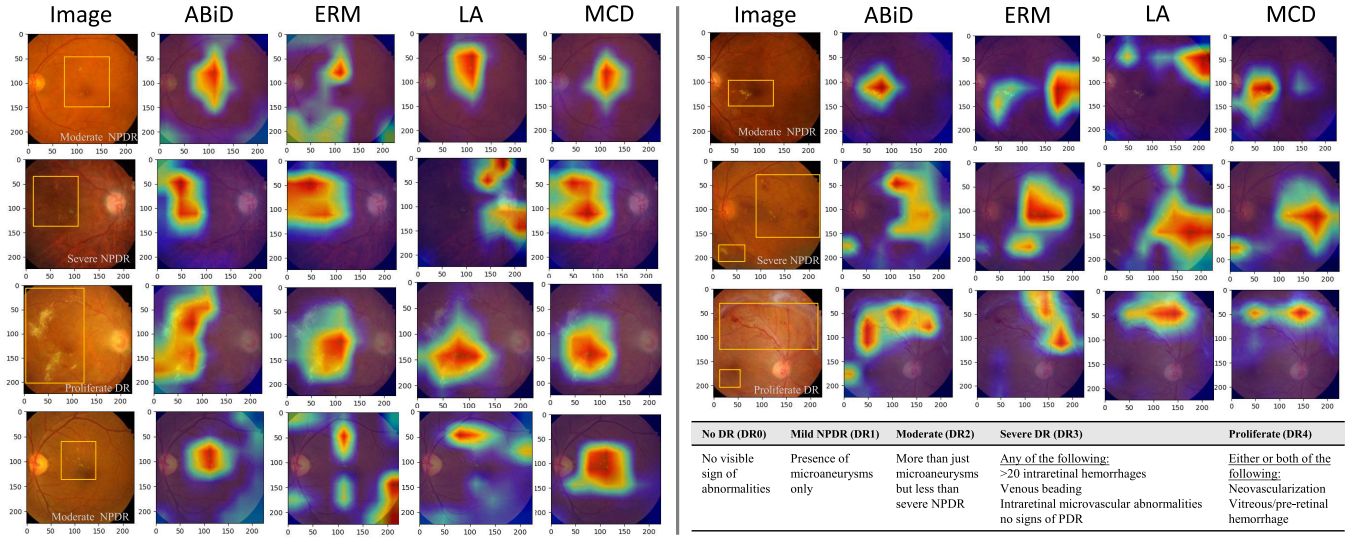


Fig. 5. CAM heatmaps resulting from the trained model using ABiD. (Bottom right) The description of each grade in the International DR Scale.

TABLE VII
RESULTS WITH DIFFERENT HYPER-PARAMETER μ

μ	Accuracy	Per-grade Accuracy	Q.W.kappa
$\mu = 0$	$60.19^{\pm 1.94}$ _	$45.59^{\pm 2.91}$ _	$68.80^{\pm 4.20}$ _
$\mu = 0.10$	$60.84^{\pm 0.65}_{\uparrow 0.7}$	$46.35^{\pm 1.84}_{\uparrow 0.8}$	$70.44^{\pm 3.82}_{\uparrow 1.6}$
$\mu = 0.20$	$60.52^{\pm 2.27}_{\uparrow 0.3}$	$48.22^{\pm 0.71}_{\uparrow 2.6}$	$72.81^{\pm 1.24}_{\uparrow 4.0}$
$\mu = 0.30$	$61.81^{\pm 1.29}_{\uparrow 1.6}$	$48.44^{\pm 2.38}_{\uparrow 2.9}$	$72.57^{\pm 1.76}_{\uparrow 3.8}$
$\mu = 0.40$	$62.78^{\pm 0.65}_{\uparrow 2.6}$	$46.94^{\pm 2.06}_{\uparrow 1.4}$	$73.54^{\pm 0.54}_{\uparrow 4.8}$
$\mu = 0.50$	$64.72^{\pm 1.29}_{\uparrow 4.5}$	$49.29^{\pm 2.83}_{\uparrow 3.7}$	$73.92^{\pm 2.48}_{\uparrow 5.1}$
$\mu = 0.75$	$63.43^{\pm 2.49}_{\uparrow 3.2}$	$48.61^{\pm 2.29}_{\uparrow 3.0}$	$73.72^{\pm 2.61}_{\uparrow 4.9}$
$\mu = 1.00$	$62.30^{\pm 2.24}_{\uparrow 2.1}$	$48.40^{\pm 2.46}_{\uparrow 2.8}$	$72.74^{\pm 3.06}_{\uparrow 3.9}$

TABLE VIII
COMBINE ABID WITH OTHER DEEP LEARNING STRATEGIES FOR DR

	IDRiD			DDR (**AVG**)		
	Acc.	PAcc.	κ.	Acc.	PAcc.	κ.
Sentry(RandAug)	54.1	49.2	66.4	63.8	46.7	61.8
ABiD+RandAug	$64.1_{\uparrow 10.}$	$52.3_{\uparrow 3.1}$	$75.9_{\uparrow 9.5}$	$74.5_{\uparrow 11.}$	$53.5_{\uparrow 6.8}$	$69.2_{\uparrow 7.4}$
RCSSL	47.9	35.6	47.5	55.7	26.2	20.2
ABiD+RCSSL	$51.1_{\uparrow 3.2}$	$38.8_{\uparrow 3.2}$	$48.6_{\uparrow 1.1}$	$61.5_{\uparrow 5.8}$	$29.6_{\uparrow 3.4}$	$26.9_{\uparrow 6.7}$
MIL-VT	57.6	44.5	68.0	70.2	45.4	62.2
ABiD+MIL-VT	$65.1_{\uparrow 7.5}$	$52.7_{\uparrow 8.2}$	$72.8_{\uparrow 4.8}$	$74.5_{\uparrow 4.2}$	$55.7_{\uparrow 10.}$	$70.1_{\uparrow 7.9}$

study on the discrepancy minimization mechanism by changing F_{times}. This step, after amplifying classifier differences to identify confusion areas, aims to minimize the variance among asymmetric classifiers, thereby compelling the model to identify essential features distinguishing between grades. The results indicate that reducing or removing the discrepancy minimization mechanism negatively impacts performance, underscoring the importance of this core mechanism.

b) Forward and inverse distribution compensation mechanism on C:: As shown in Table VI (line 8-10), we ablation the times of the forward and inverse distribution compensation mechanism to understand the impact of this component. Reducing the frequency diminishes the distinction between grading boundaries, which negatively affects model performance. Additionally, we compared with the original maximum discrepancy on the target data mechanism. We found that the results were even worse than without this part. This phenomenon is because the target data's pseudo-labels are unreliable under the grade-skewed scenarios.

Table VII shows results for different μ values. Smaller μ values lead to less classifier discrepancy and reduced model effect. Larger μ values do not consistently improve performance. This is because, as μ increases, the proportion of easily confused points decreases, resulting in a decline in error rates due to distribution compensation. However, even with high μ values, performance remains superior compared to scenarios without ABiD method.

2) Combine With Other Augmentation Strategies: Note that ABiD can be combined with any data augmentation strategy, that is, $\mathcal{M}(\cdot)$ is flexibly replaceable. We replaced the original $\mathcal{M}(\cdot)$ with the RandAug [37] strategy used in SENTRY and show results in Table VIII. Concretely, RandAug strategy proposed a greatly simplified search space for data augmentation with 2 hyper-parameters: the number of augmentation transformations N and transformation magnitude M. Following [21], we set $N = 3$ and $M = 2$. Using the same augmentation, our method remained significantly superior to SENTRY.

3) Combine With Other Deep Learning Strategies: On the other hand, the backbone network of ABiD method can also be replaced flexibly. The ABiD method can build upon these specialized DR methods to further enhance their performance. We introduce several backbone networks specifically designed for fundus images: Rotation-Oriented Collaborative Self-Supervised Learning for Retinal Disease Diagnosis (RCSSL) [42] and Multiple instance learning enhanced vision transformer (MIL-VT) [43]. We use the original fine-tuning method and the proposed ABiD domain adaptation method on these backbone networks, respectively. As shown in Table VIII, the results demonstrate that integrating the ABiD method with these approaches consistently enhances model performance in comparison to traditional fine-tuning methods. Additionally, Table IX shows the performance of the ConvNeXt-Tiny model with an input size of 384 × 384, demonstrating a clear improvement. This enhancement is due

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TABLE IX
RESULTS OF METHOD ABID‡ WITH DIFFERENT INPUT SIZES

	IDRiD			DDR (**AVG**)		
input size	Acc.	PAcc.	κ.	Acc.	PAcc.	κ.
224×224	65.70	52.65	76.80	75.56	53.21	69.89
384×384	69.26$_{\uparrow 3.6}$	56.45$_{\uparrow 3.8}$	79.78$_{\uparrow 3.0}$	78.07$_{\uparrow 2.5}$	57.49$_{\uparrow 4.3}$	74.29$_{\uparrow 4.4}$

TABLE X
TUNING AND TESTING TIME (SECONDS) ON IDRID DATASET

	testing time	tuning time / epoch	Acc. (↑)
SENTRY‡	1.601	1.505	54.05
PAT‡	1.924	1.583	59.55
ABiD‡ (time = 1)	1.416	1.178	63.11
ABiD‡ (time = 2)	1.422	1.871	65.70

to the larger input size capturing finer details, indicating that our method can further improve with larger input sizes.

4) Visualization: Conducting an examination of the Class Activation Mapping (CAM) [44] heatmaps derived from the image-level supervised model using the ABiD method (refer to Fig. 5). It has come to our attention that these heatmaps provide valuable insights into the specific areas of focus where lesions occur within the images. By comparing the description of DR Grade in the International DR Scale, we demonstrate that ABiD effectively targets the key lesions through image-level supervised training.

Moreover, we conducted comparisons with other methods, including ERM, LA and MCD. In many instances of these methods, the highlighted regions did not correspond accurately to the lesion locations. This discrepancy arises because these models tend to memorize individual characteristics rather than the fundamental features across different disease grades. MCD outperformed the other comparison methods but still showed incomplete lesion detection (e.g., partial identification of hemorrhagic spots). This underscores the limitation of traditional dual-classifiers in detecting small inter-class lesion differences. In contrast, the ABiD method overcame this challenge and demonstrated excellent performance.

5) Computational Resource Analysis: For spatial complexity, the introduced asymmetric classifier has a small impact. The C_{BO} classifier added just 1,285 parameters to the 28M parameters of ConvNext-Tiny, making its impact negligible. Regarding time efficiency, as shown in Table X, ABiD‡ exhibits a clear advantage in test time. Training time correlates with the number of adversarial iterations. ABiD‡ (time=1) achieves the shortest training time per epoch at 1.178s, outperforming SENTRY and PAT. While adversarial time increases, ABiD‡ (time=2) boosts accuracy from 63.11% to 65.70%.

V. DISCUSSION

In real-world applications of image-level DR grading, domain shifts caused by cross-center, cross-user, and cross-device variations often result in poor model performance on test data. Moreover, due to the inherent characteristics of DR grading, there are issues of grade skew and grade distribution shifts across different datasets. We refer to these challenges collectively as the grade-skewed domain adaptation (GSDA) problem and propose the first method, ABiD, to address it. We introduce the forward and inverse distribution compensation mechanism to enhance the difference between the dual classifiers near the decision boundaries, identifying confusing instances. This method forces the feature extractor to generate features with minimized discrepancies. We conducted 21 adaptation tasks on public and private datasets, with domain splits closely resembling real-world scenarios. Extensive experiments demonstrate that our method effectively handles GSDA problem in retinal fundus images.

Despite the favorable performance achieved by our method, there are inherent limitations that should be further considered. First, a significant limitation pertains to real-time applications. The domain adaptation process requires model training, which demands computational resources. A potential solution is to pre-train the model using a few target domain samples and deploy the adapted model to the unseen target data. This one-step adaptation allows the model to fit better target data than training solely on the source domain, providing on-the-fly results without backpropagation during testing. In the future, balancing adaptation performance with computational demands will be a critical research direction.

Another limitation arises from the fact that, while our method demonstrates strong performance in identifying small lesions, training with image-level supervision presents challenges in accurately detecting lesions compared to pixel-level supervision. On the other hand, pixel-level supervision requires more computational resources and demands high-quality pixel-level annotations, increasing the difficulty of deploying in real-world scenarios. To improve the detection of small lesions with image-level supervision, a larger input size can help by capturing finer details. Additionally, the ABiD paradigm can be effectively combined with other data augmentation methods and more powerful backbone networks to further boost its performance. Balancing model performance with practical considerations remains an important challenge.

Our ABiD method is also feasible for the expansion to the grading of additional correlated diseases with GSDA problem. In this paper, we focus on DR grading, but our method could also be applied more broadly. We also plan to investigate the GSDA problem in other medical scenarios, such as Diabetic macular edema (DME) grading, Age-related macular degeneration (AMD) grading, mammography BI RADS grading, and cancer histopathology grading, facilitating improved domain adaptation of models on the target domain. Through the forward and inverse distribution compensation mechanism, our method can better locate similar features of adjacent grades of diseases. The target samples near classification boundaries or away from source support are separated by minimizing ABiD discrepancy so that they are correctly classified.

It is also a very interesting direction how to trade off the classifier difference amplification mechanism under the broader source domain label distribution. In the future, we are focused on advancing our research to enhance the adaptive capabilities of models in changing more complex real-world medical applications.

VI. CONCLUSION

In this paper, we study an unexplored but common problem in the medical scene, namely grade-skewed domain adaptation (GSDA). Based on this challenging setting, for diabetic

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retinopathy grading, we provide a novel methodology Asymmetric Bi-Classifier Discrepancy Minimization via introducing forward and reverse compensation based on skewed source distribution, the divergence between the bi-classifier near the decision boundary is enhanced to find confusing instances of the classification boundary, and then the feature extractor is forced to generate features that minimize the difference. Extensive experiments based on one toy dataset and three real-world datasets demonstrate significant advantages of ABiD over previous state-of-the-art methods.

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